Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/10/2005
PI Name:	Hoel, David G Ph.D.		
Project Title:	The Analysis Of The Potential Health Risks From High LET Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	29425-2203	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
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No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Carnes, Bruce (University of Oklahoma)		
Grant/Contract No.:	NAG9-1518		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Extensive pathology and mortality data from Argonne National Laboratory for B6CF1 mice receiving single and protracted exposure to neutrons will be used to estimate radiation-induced risk (mortality and morbidity) for heavy ions. The justifications for using neutron data as a surrogate for HZE particles are that the 0.85 MeV JANUS neutrons are a high LET radiation, and their biological effectiveness and dose-response at lower doses have been shown to be similar to those for Fe and Argon. Although we will acquire heavy ion mortality data as it becomes available, the neutron data will be used to fit both empirical and biologically motivated dose-response models. Empirical models will be examined because they have already been applied to these data in order to successfully make interspecies predictions (mouse to dog to human) of radiation-induced mortality for low LET radiation (both acute and protracted). The biological models will be examined to not only compare their predictions to the empirical models, but also to examine how knowledge of the pathogenesis of disease processes affects quantitative models used to predict radiation-induced mortality risks. Finally, the results of this modeling project will be synthesized into practical estimates of the potential radiation risks for		

	astronauts on prolonged space flights. To obtain neutron RBE values for a variety of cancer and cardiovascular endpoints. Evaluate the assumption of a constant RBE over dose levels. Examine the predictability of gamma exposed animals to human cancer risk. Will provide needed information for estimating astronaut cancer risks from HZE exposures in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Help to set exposure limits for workers exposed to high LET radiation.
Task Progress:	The initial phase of the project has been focused on the identification of relevant data from the animal database at Argonne National Laboratory (ANL) and appropriate pathology endpoints for analysis. Upon review, seven end-points identified histopathologically as having either a specific origin or population of cells at risk have been identified for this project: lung tumors, liver tumors, and Harderian gland tumors representing the epithelial tissues; lymphocytic-lymphoblastic sarcoma and angiosarcoma representing the connective tissues; mammary tumors and granulosa cell ovarian tumors representing the female reproductive system. We also are examining radiation effects on CVD mortality. Results Total mortality provides the best data due to sample sizes. Figures 1 &2 show the results for the relative risk calculated use Cox proportional hazard models for each dose group. Restricting the data to < 0.76 Gy for neutron and <2.6 Gy for gamma we estimate a neutron RBE of 8.4 and 10.8 for females and males, respectively. The two most frequent cancers are lung and lymphoma (Figures 3 & 4). For lung cancer the low dose neutron (<0.1Gy) and gamma (<0.5Gy) had relative risks less than one among the female mice which have a low background compared to males (7% vs. 29%). This was not seen in the males but they did not have the low dose exposures as did the female mice. A similar pattern is seen for lymphomas . The preliminary RBE values for the chosen cancer endpoints mentioned above and will need careful individual analysis in order to deal with non-linearities. (note: I was totally unsucessful in finding a way to paste figures in this online document. If needed they could be sent separately) TABLE Relative Risks and estimated RBE values from acute exposure expiments. Gamma < 2.5 Gy and Neutron < 0.75 Gy FEMALE MALE Gamma Neutron RBE Gamma Neutron RBE Mortality 1.40 4.36 8.39 1.12 2.29 10.8 Lung 2.52 26.12 16.50 1.05 2.71 33.4 Hardarian 5.69 72.11 15.17 3.67 20.28 7.2 Lymphoma 1.95 17.17 17.11 1.31 2.06 3.5 Angiosarcoma 2.34 17.36 12.18 1.46 2.75 3.8 Liver 3.42 81.80 33.39 1.34 3.75 8.0 Mammary 1.83 9.84 10.71 Ovary 2.77 74.14 41.23 An important aspect of the neutron RBE estimates is their uncertainty. Simulation metions have been used in a newly submitted manuscript (Priest, Hoel & Brooks 2005) involving alpha RBEs and it will be applied to the estimated Argonne cancer RBE estimates. A manuscript involving these cancer endpoints for both acute and fractionated exposures is being developed. From the A-bomb studies in Japan increased mortality from CVD has been observed. The data from the Argonne Janus program had CVD contributing to mortality depending on the experiment (i.e. pathologist). This consisted of increases in endocarditis, myocarditis and panarteritis. We have carried out a considerable amount of analysis of this data and find that there is a low dose effect for neutrons. Also the RBE for neutrons appears to range from about 2.5 to 5. Neutron Gamma RBE Experiment exposures sex ERR (s.e.)* ERR (s.e.) JM-2 1 F 2.23 (0.47) 0.77 (0.19) 2.9 M 0.91 (0.26) 0.19(0.11) 4.9 JM-3 1 M 3.95 (2.31) no effect JM-9 1 F 15.0 (12.6) 5.65 (5.38) 2.7 JM-13 60 F 7.79 (5.53) 1.88 (1.49) 4.2 M 1.76(2.17) 0.16(0.37)?? A manuscript is being written which addresses both the issue of low-dose CVD risk and neutron RBE. Two additional activities concern analysis tool development. First, we have a working 2-stage clonal expansion model for cancer risk and will use it in the cancer analysis after finishing the above mentioned manuscript. Second, we are experimenting with Bayesian models for estimating RBE uncertainty. This, if successful, will result in a publication. Completed manuscripts and publications supported by this project are as follows. Hoel, DG, Carnes B. Cancer Dose-Response Analysis of the Radium Dial Workers. Proceedings from the HEIR 2004 9th International Conference on Health Effects of Incorporated Radionuclides - Emphasis on Radium, Thorium, Uranium and their Daughter Products. GSF-National Research Center, Neuherberg, Germany, November 29-December 1, 2004. 2005 (in press) Priest ND, Hoel DG and Brooks PN. Relative Biological Effectiveness (RBE) of chronic irradiation by 45Ca ß-particles and 242 Cm a-particles with respect to the production of lung tumours in CBA/Ca mice. International Journal of Radiation Biology, 2005 (submitted). Carnes, B.A., and S.J. Olshansky. 2004. Intrinsic mortality: a bridge between biology and mortality analysis. Journal of Gerontology: Biological Sciences (resubmitted, pending).
Bibliography Type:	Description: (Last Updated: 02/25/2014)
Abstracts for Journals and Proceedings	Hoel, D.G.; Carnes, B. " Dose-Response Analysis of the Radium Dial Workers" Proceedings from the HEIR 2004 9th International Conference on Health Effects of Incorporated Radionuclides - Emphasis on Radium, Thorium, Uranium and their Daughter Products. GSF-National Research Center, Neuherberg, Germany, November 29-December 1, 2004. 2005 (in press) unknown , Nov-2005